

## HW3

$$1. (a) [a x_1(t-2) + b x_2(t-2)] + [a x_1(2-t) + b x_2(2-t)]$$

$$= a[x_1(t-2) + x_1(2-t)] + b[x_2(t-2) + x_2(2-t)]$$

$$T(x(t-t_0)) = x(t-t_0-2) + x(2-t-t_0)$$

$$\neq x(t-t_0-2) + x(2-t+t_0)$$

$\therefore$  time-variant; linear

$$(b) \cos(3t)[a x_1(t) + b x_2(t)]$$

$$= a \cos 3t x_1(t) + b \cos 3t x_2(t)$$

$$\cos 3(t-t_0)x(t-t_0) \neq \cos 3t x(t-t_0)$$

$\therefore$  time-variant; linear

$$(c) \int_{-\infty}^{2t} a x_1(\tau) d\tau + b x_2(\tau) d\tau$$

$$= a \int_{-\infty}^{2t} x_1(\tau) d\tau + b \int_{-\infty}^{2t} x_2(\tau) d\tau$$

$$\int_{-\infty}^{2t} x(\tau-t_0) d\tau \neq \int_{-\infty}^{2(t-t_0)} x(\tau) d\tau$$

$\therefore$  time-variant; linear

(d) when  $t < 0$ , it is linear absolutely

$$\text{when } t > 0 \quad a x_1(t) + b x_1(t-2) + a x_2(t) + b x_2(t-2)$$

$$= a[x_1(t) + x_2(t)] + b[x_1(t-2) + x_2(t-2)]$$

$\therefore$  time-variant; linear

(e) when  $x(t_1), x(t_2) < 0$   $y(t_1) + y(t_2) = 0$

$y(t_1+t_2)$  might not equal to 0

$$\int_0^{\infty} 0, x(t) < 0 = \int_0^{\infty} 0, x(t) < 0$$

$$\int_0^{\infty} x(t-t_0) + x(t-t_0-2) = \int_0^{\infty} x(t-t_0) + x(t-t_0-2)$$

$\therefore$  no linear; time-invariant

$$(f) a x_1\left(\frac{t}{3}\right) + b x_2\left(\frac{t}{3}\right)$$

$$x\left(\frac{t-t_0}{3}\right) \neq x\left(\frac{t}{3}-t_0\right)$$

time-variant; linear

$$(g) \frac{d(a x_1(t) + b x_2(t))}{dt} = a \frac{dx_1(t)}{dt} + b \frac{dx_2(t)}{dt}$$

$$\frac{dx(t-t_0)}{dt} = \frac{dx(t-t_0)}{d(t-t_0)}$$

$\therefore$  time-invariant; linear

$$2. x(t) = u(t) + u(t-1) - u(t-2) - u(t-3)$$

$$y(t) = \begin{cases} 0 & , t \in (-\infty, 0) \cup (2, +\infty) \\ t & , t \in (0, 1) \\ t-1 & , t \in (1, 2) \end{cases}$$

$$y(t) = s(t) + s(t-1) - s(t-2) - s(t-3)$$

$$\textcircled{1} t \in (-\infty, 0) \quad 0 = 0$$

$$\textcircled{2} t \in (0, 1) \quad s(t) = t$$

$$\textcircled{3} t \in (1, 2) \quad s(t) + \underbrace{s(t-1)}_{u \in (0, 1)} = t-1$$

$$s(t) + t-1 = t-1 \Rightarrow s(t) = 0$$

$$\textcircled{3} t \in (2, 3) \quad s(t) + s(t-1) - s(t-2) = 0$$

$$s(t) + 0 - t+2 = 0$$

$$s(t) = t-2$$

$$\textcircled{4} t \in (3, 4) \quad s(t) + s(t-1) - s(t-2) - s(t-3) = 0$$

$$s(t) + t-3 - 0 - t+3 = 0$$

$$s(t) = 0$$

$$\textcircled{5} t \in (4, 5) \quad s(t) + s(t-1) = s(t-2) + s(t-3)$$

$$s(t) + 0 = t-4 + 0$$

$\therefore$  there are two terms in

$$s(t) + s(t-1) = s(t-2) + s(t-3) \text{ will be } 0$$

$$\therefore s(t) = \begin{cases} t-2k & , t \in (2k, 2k+1) \\ 0 & , t \in (2k+1, 2k+2) \end{cases}$$

3.

$$(a) \quad t < 0 \quad e^{-4t} u(t-2) = 0$$

$$\int_{-\infty}^{\infty} e^{-4t} u(t-2) dt = \int_2^{\infty} e^{-4t} dt = -\frac{1}{4} e^{-4t} \Big|_2^{\infty} < \infty$$

causal and stable

$$(b) \quad t < 0 \quad e^{-bt} u(3-t) \neq 0 \quad \lim_{t \rightarrow -\infty} e^{-bt} = \infty$$

$\therefore$  non-causal and non-stable

$$(c) \quad t \in (-50, 0) \quad h(t) \neq 0 \quad \int_{-\infty}^{\infty} e^{-2t} u(t+50) dt = -\frac{1}{2} e^{-2t} \Big|_{-50}^{\infty} < \infty$$

$\therefore$  non-causal & stable

$$(d) \quad t < 0 \quad h(t) \neq 0 \quad \int_{-\infty}^{\infty} h(t) dt = \frac{1}{2} e^{2t} \Big|_{-\infty}^{-1} < \infty$$

non-causal & stable

$$(e) \quad t < 0 \quad h(t) \neq 0 \quad \int_{-\infty}^{\infty} h(t) dt = 2 \int_0^{\infty} e^{-bt} dt = 2 \times \left(-\frac{1}{b}\right) e^{-bt} \Big|_0^{\infty} < \infty$$

non-causal & stable

$$(f) \quad t < 0 \quad h(t) = 0$$

$$\int_{-\infty}^{\infty} t e^{-t} u(t) dt = \int_0^{\infty} t e^{-t} dt = (-t-1) e^{-t} \Big|_0^{\infty} = 1 < \infty$$

causal & stable